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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (VDE) (CBCS) Degree Examinations
January 2023

22VDE101 - DIGITAL VLSI DESIGN

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

Unit – I

Marks BT* CO* PO*

1. a) Draw the energy band diagram of a MOS system in inversion. Using the measured data given for a NMOSFET, determine V_{T0} and μ_n assuming $|2\phi_F| = -0.64$ V, $(W/L)=1$ and $t_{ox} = 345$ Å.

V_{GS} (V)	V_{DS} (V)	$I_D(\mu A)$
4	4	256
5	5	441
4	4	144
5	5	256

- b) Derive the expression for V_{OH} and V_{IL} of a resistive load inverter.

10	L*3	1	1
10	L2	1	1

2. a) With necessary derivations, calculate V_{OH} and V_{OL} in a resistive load inverter given that $V_{DD}=5$ V, $k_n' = 20 \mu A/V^2$, $V_{T0} = 0.8$ V, $R_L = 200$ k Ω and $(W/L)_n = 2$.

10	L3	1	1
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- b) Write constant field scaling and constant voltage scaling models. Also determine the effect on oxide capacitance, drain current in linear and saturation region and power dissipation using both models.

10	L2	1	1
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Unit – II

3. a) For a 3-stage ring oscillator, plot the output waveforms and derive the expression for frequency of oscillation. Also design a ring oscillator to generate a frequency of 100 MHz, assume that average propagation time of an inverter is 1 ns.

10	L2	2	3
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- b) Calculate noise margin NM_L in a CMOS inverter with the following parameters: $V_{DD} = 3.3$ V, $V_{T0,n} = 0.6$ V, $V_{T0,p} = -0.7$ V, $k_n = 200 \mu A/V^2$ and $k_p = 80 \mu A/V^2$. Derive necessary equations.

10	L3	2	3
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4. a) With necessary derivation, analyze the dependence of power dissipation in a CMOS inverter on the signal switching speed and the power supply.

10	L2	2	3
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- b) Using differential equation method, calculate the delay time for the output of a CMOS inverter to fall from an initial value of 3.3 V to 1.65 V for a step pulse input which switches instantaneously from 0 V to 3.3 V. Given: $V_{DD} = 3.3$ V, $V_{T,n} = 0.8$ V, $C_{load} = 300$ fF., $I_{sat} = 2$ mA with $V_{GS} = 3.3$ V and $V_{DS} > 2.5$ V,

10	L3	2	3
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P.T.O.

Unit – III

5. a) Explain the concept of multiple output domino logic. Also design a multiple output domino CMOS gate to realize four outputs C_1, C_2, C_3 and C_4 defined by,
- $$C_1 = G_1 + P_1 C_0 ; \quad C_2 = G_2 + P_2 G_1 + P_2 P_1 C_0$$
- $$C_3 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 C_0$$
- $$C_4 = G_4 + P_4 G_3 + P_4 P_3 G_2 + P_4 P_3 P_2 G_1 + P_4 P_3 P_2 P_1 C_0$$
- b) Explain the dynamic voltage bootstrapping technique to overcome threshold voltage loss in digital circuits.
6. a) Illustrate cascading problem in dynamic CMOS logic. Design static CMOS circuit and dynamic CMOS circuit using minimum number of MOSFETs for the gate defined by $F6 = AB + (C + D)(E + F) + GH$.
- b) Explain Domino CMOS logic. How charge sharing problem can affect the circuit output and how it can be prevented? Explain.

Unit – IV

7. a) Explain array multiplication of $X_3 X_2 X_1 X_0$ with $Y_3 Y_2 Y_1 Y_0$. Hence develop the structure of 4 X 4 array multiplier.
- b) i) Construct adder/subtractor using XOR gates and adders
ii) Draw the logic diagram for zero and one detectors.
8. a) Design a 4X4 barrel shifter using transmission gates and explain the arithmetic and logical shift operations.
- b) Implement comparator using i) Adders ii) XNOR gates iii) Pass gates.

Unit – V

9. a) What is the motivation for design for manufacturability? How process variations affect MOSFET drain current. Suggest a method to make the circuit performance less sensitive to process variations.
- b) With relevant diagrams, explain how the parasitic BJTs in CMOS structures leads to latch up. Also state the remedies for the same.
10. a) Explain transmission gate based input circuit and tristable output circuit used in ICs.
- b) Design a clock decoder circuit which generates four clock phases assuming two primary clock signals. Draw the H-tree structure used for on-chip clock distribution with uniform delay.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
